

NaijaReview AI – Task A: User Modelling for Review Simulation using LLM

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Abstract

The abstract user modeling component of **NaijaReview AI**, an LLM-powered agentic system that models users as dynamic, context-sensitive individuals and simulates their reviews for unseen items. Our approach combines SVD collaborative filtering with Bayesian shrinkage for rating prediction (RMSE 0.956) and a vocabulary-forced LLM generation strategy for behavioral fidelity (BERTScore 0.901, ROUGE-1 0.339). A key contribution is our **vocabulary-forcing technique**: on datasets where 68% of reviews contain fewer than 15 words, constraining the LLM to reuse the user's exact lexicon yields a +21% ROUGE improvement over baseline few-shot prompting. We also report a cautionary finding on Team Cerebral DSN × BCT Data & AI Summit Hackathon 3.0 – May 2026

Keywords: LLM agents, user modeling, review simulation, collaborative filtering, SVD, vocabulary forcing, Nigerian NLP

1. Introduction We are trained on data up to October 2023.

Online review platforms are rich sources of human behavioral data, but most AI systems treat users as static profiles. In Task A, we are asked to develop an agent that, given a user's review history ($N-1$ reviews), can predict the star rating and the written review for a held-out unseen item, including tone, vocabulary, and contextual nuance.

There are three challenges here: (1) reviews are very personal, and two users reviewing the same product will use different styles and registers, (2) most reviews are short (median length 9 words), so every generated word counts towards evaluation metrics, and (3) the system must generalize to unseen items without any item-specific training data.

Our main contributions:

1. **SVD + Bayesian bias rating prediction** that outperforms XGBoost at inference due to feature distribution alignment
 2. **The XGBoost distribution mismatch finding**—strong training RMSE (0.613) does not guarantee inference performance (1.193) when feature distributions shift
 3. **Vocabulary-forcing for short-form reviews** yielding +21% ROUGE-1 improvement
 4. **Nigerian contextualization, adaptation of Pidgin English and modeling of cultural tone**
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2. System Architecture

NaijaReview AI follows a modular, agent-based architecture:

- **Persona Builder Agent** — writing style analysis (length, vocabulary richness, punctuation)
2. Architecture of the Systemication, rating distribution statistics, category preference extraction
- **Rating Predictor Agent** — **Bayesian shrinkage bias (weight 0.6), SVD collaborative filtering (weight 0.4), category interaction blend (optional).**
- **Review Generator Agent** — vocabulary forcing (must-use word list), exact phrase
Review Generator Agent - vocabulary forcing (must use word list), exact phrase templates, tone + length enforcement

LLM Strategy: Groq (Llama-3.3-70B-Versatile) as primary and OpenAI GPT-4o-mini as fallback. After a single Groq failure (typically at the 100K daily token limit), all subsequent LLM strategies are Groq (Llama-3.3-70B-Versatile) as the main, falling back to OpenAI GPT-4o mini. One failed Groq call (usually at the 100K daily limit of tokens) sends all calls to OpenAI—a circuit-breaker pattern that removes $\sim 3\times$ latency from repeated failed requests.

Memory Store: JSON-backed persistent user profiles cache persona analyses and styles.
Memory Store: Caches persona analyses and style fingerprints as persistent user profiles backed by JSON to save $\sim 40\%$ in tokens.

3. Method of Study

3.1 Persona Development

The PersonaBuilder extracts a behavioural fingerprint from a user's $N-1$ review history: **writing style metrics** (average length, vocabulary richness, punctuation density,

capitalization), tone classification (LLM-classified, e.g., "harsh critic" or "enthusiastic recommender"), rating distribution (mean, std, skewness), and category preferences (top categories and category-specific rating tendencies). Personas are calculated once and cached.

3.2 Rating Prediction: SVD + Bayesian Bias

The XGBoost Distribution Mismatch We trained our first XGBoost model on 22 engineered features, achieving 0.613 RMSE in training. However, at inference, it gets averaged user-history features instead of per-review features—a fundamentally different distribution, resulting in a ballooning RMSE of 1.193. This is a generalizable lesson: strong training metrics do not guarantee inference performance when feature distributions change between training (per-review) and inference (user-level aggregation).

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1. **Bayesian Shrinkage Bias (weight 0.6):** $\text{predicted_rating} = \text{global_mean} + \text{user_bias} + \text{item_bias}$, with biases shrunk toward the global mean ($\lambda=10$) to prevent overfitting for sparse users/items.
2. **SVD Collaborative Filtering (weight 0.4):** Truncated SVD with 50 latent factors on a sparse user-item residual matrix (4,500 users $\times$ 7,000 items), capturing interaction patterns beyond simple biases.
3. **Category Interaction (optional, 20% blend):** When a user has ≥ 2 ratings in the target item's category, their category-specific mean is blended at 20% weight.

3.3 Review Generation: Vocabulary-Forcing

The Short-Review Problem. 67.9% of reviews contain fewer than 15 words (median 9). This transforms ROUGE into a *precision* problem—a single additional word match shifts ROUGE-1 by ~ 0.11 . This insight motivated vocabulary forcing: vocabulary selection, not fluency, is the primary lever for quality in short reviews.

The Vocabulary-Forcing Technique:

1. **Must-Use Word List:** Top 8 most frequent content words (excluding stop words) from the user's history. The prompt mandates ≥ 4 appear in the generated review.
2. **Exact Phrase Templates:** For users averaging 4–15 words, short reviews from their history serve as copy templates with appropriate substitution.
3. **Same-Category Prioritization:** Reviews from matching categories become primary templates, capturing domain-specific vocabulary (e.g., "battery life" for electronics).

Behavioral Fidelity Enforcement: The LLM cannot override the predicted rating; output is truncated at sentence boundaries if it exceeds 1.5× the user’s average length, and the classified tone of the persona is injected into the system prompt.

4. Contextualization in Nigeria

Language Adaptation: System prompts detect Nigerian users and instruct the LLM to adopt Nigerian English naturally. If pidgin expressions are included, they are included if appropriate, but the system keeps the user's own register. Regional variations are modeled for Lagos, Abuja, and Port Harcourt.

Persona Archetypes: Out-of-the-box archetypes boost generation: “Lagos Professional” (quality-driven, formal), “University Student” (price-sensitive, casual with slang), “Market Trader” (pragmatic, heavy in Pidgin)

Cultural References: Generated reviews reference local realities (NEPA/power outages, local brands like Oraimo) where contextually appropriate.

Mode	Car Reviews
Nigeria	“This power bank is too good sha, e dey charge quick quick. Great for when NEPA takes light.
Generic drugs	Great power bank, fast charging. Great for power outages.”

5. Experiments and Ablation Study

5.1 Dataset and Evaluation

We evaluate **59,981 reviews** (4,500 users, 7,000 items) from Amazon Electronics, Yelp, and Goodreads. For evaluation, **30 users** with diverse histories are selected; each user's last review is held out as ground truth. Metrics: **RMSE** (rating accuracy) and **ROUGE-1** (unigram 5.1 dataset and evaluation).

5.2 Ablation Study

System type	RMSE	ROUGE-1.	BERTScore
Baseline (Bayesian bias + generic few-shot LLM)	0.945	0.280	0.850
+ Promptings of category pairing	0.945	0.324	0.903
+ XGBoost 70/30 blend (did not work)	1.193	0.297	0.878
* Back to pure Bayesian bias	0.953	0.319	0.900
+ SVD collaborative filtering (50 components)	0.956	0.319	0.900
Final: +forcing vocabulary +templates	0.956	0.339	0.901

Prompting Techniques	ROUGE-1.	Get
Few-shot generic only	0.280	standard
+ Category match	0.324	+15.7%
* Vocabulary pushing * templates	0.339	+21.1%

5.3 Major findings

1. **Vocabulary forcing is the dominant lever:** +21% ROUGE-1 validates that vocabulary selection matters more than fluency for short reviews.
2. **XGBoost distribution mismatch is generalizable:** Despite 0.613 training RMSE of 0.613, inference RMSE of 1.193 demonstrates the danger of feature distribution shift.
3. **SVD stabilizes predictions:** The slight RMSE increase (0.953 → 0.956) is offset by more robust cold-start predictions.
4. **Outlier users dominate RMSE:** Four extreme users account for ~0.35 RMSE points; excluding them yields effective RMSE ~0.60.
5. **BERTScore is robust:** Semantic similarity remains high (0.850–0.903) across all variants.

6. Qualitative Example

User profile: 8 reviews, avg rating 3.8, tone: "casual conversationalist," avg length: 11 words.
Target item: Samsung Galaxy Buds3 Pro.

	Text 3.	Scores
Real Data	“Sound quality is awesome. Fits good. Worth it.	4
When the wind blew, it would make a noise in the chimneys, and the flames of the fire would dance and flash.	Great sound quality, comfortable fit. Good product, worth it.	4.02

The generated review reuses 6 of 9 ground-truth words (ROUGE-1: 0.67), matching the user’s terse style—showing reconstruction of individual voice, not generic output.

7. Technical Stack & Reproducibility

Part 1	Tech 1.
LLM’s	Template for Rating
Rating Model	SVD (50 factors) + Bayesian shrinkage ($\lambda=10$)
Persona Engine	Persistent profile cache backed by JSON
Memory	JSON-backed persistent profile cache
Stack Overflow	docker-compose, FastAPI + Streamlit

Deployment: docker-compose up --build runs the API (port 8000) and UI (port 8501).
Reproducibility: We release all the preprocessing, training, and evaluation scripts. SVD uses fixed random seeds. LLM outputs are cached to allow deterministic replay.

8. Future Work

- **Rating (RMSE -> 0.75) Neural Collaborative Filtering (NeuMF) Item content embeddings Ensemble with LightGBM with correct feature alignment at inference time Review (ROUGE→0.45): retrieval-augmented generation from similar past reviews user-specific few-shot tuning character-level style transfer evaluation: Statistical significance with 100+ user cohort Human evaluation for naturalness**

- **Review (ROUGE→0.45): retrieval-augmented generation from similar past reviews user-specific few-shot tuning, character-level style transfer**
 - **Evaluation: 100+ user cohort for statistical significance; human evaluation for naturalness**
 - **Evaluation: Statistical significance with 100+ user cohort Human evaluation for Nigerian NLP: Real-time Pidgin detection; Yoruba, Igbo and Hausa code-switching**
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9. CONCLUSIONS

According to our user modeling system, hybrid architectures that integrate statistical methods and LLM reasoning achieve strong performance in review simulation. Key contributions: (1) an XGBoost distribution mismatch finding—a generalizable lesson for aggregate-level inference; (2) vocabulary forcing yielding a +21% ROUGE-1 for short reviews; and (3) persona-driven generation capturing individual users' voices. The Nigerian case study demonstrates how LLM agents can be culturally adapted without specific training data.

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